Karan Singh Rana

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EDUCATION

University of Texas at Dallas Expected May 2029
Bachelor of Science in Computer Science | GPA: 3.0+
Relevant Coursework: Introduction to Programming (C++), Calculus I, Rhetoric & Composition II

ACADEMIC RESEARCH WORK

Research Paper: AI Synthetic Data Generation & Model Collapse *Rhetoric 1302, UTD | Spring 2025*

- Authored an academic research paper investigating synthetic data generation in large language models and the phenomenon of model collapse
- Analyzed how training AI models on AI-generated data degrades model performance over successive generations, with implications for long-term LLM reliability
- Engaged with current ML research literature to synthesize findings and construct an evidence-based argument

TECHNICAL SKILLS

Languages: Python, C++, Java
Tools & Software: Microsoft Excel, Microsoft Word, Git, React
Certifications: CodeHS Cybersecurity Certificate, CompTIA Computer Hardware Certificate

PROJECTS & EXPERIENCE

Notebook Web Application *Nebula Labs, UTD*

- Built a full-stack web application using React as part of Nebula Labs, UTD's student-run open source organization
- Contributed to front-end development, implementing UI components and user interaction flows
- Collaborated with a team of student developers following real-world software development practices

ACTIVITIES & LEADERSHIP

President, Esports Club *Princeton High School*

- Founded and led the school's esports club, managing scheduling, team coordination, and competitive events
- Coached teammates in Apex Legends strategy, communication, and performance under pressure

Competitive Gaming — Apex Legends

- Competed at a high level in Apex Legends, developing skills in strategic decision-making, rapid adaptation, and performance under pressure

Badminton

- Recreational competitive player with consistent training and match participation

HONORS & AWARDS

- AES Scholarship Recipient
- Princeton High School (Texas) — Honors Graduate
- CompTIA Computer Hardware Certificate
- CodeHS Cybersecurity Certificate